Lecture 10 recap

In Lecture 10:

- Introduced improper integrals and learnt how to assess their convergence
- Defined the Laplace transform and discussed its linear properties
- Set out criteria for existence of the Laplace transform of a function

Lecture 11

In Lecture 11, we will:

1. Investigate existence of the Laplace transform for example functions
2. Introduce the inverse Laplace transform
3. Use the LT "cheat sheet" to calculate LT and ILT in examples

Exercise 3.6: Do Laplace transforms exist for the following functions?

$$f_1(t) = t^3\, e^{2t}\,, \qquad f_2(t) = e^{t^2}$$

Let's examine if e^{3t} is greater than $f_1 = t^3 e^{2t}$ for large t.

$$\lim_{t\to\infty} \frac{t^3 e^{2t}}{e^{3t}} = \lim_{t\to\infty} t^3 e^{-t} = 0$$

There will be some $T > 0$ at which e^{3t} becomes greater than $t^3 e^{2t}$.
Therefore, f_1 is bounded by an exponential and must have a Laplace transform.

f_2 grows faster than any exponential $k\, e^{\alpha\, t}$ and so does not have a Laplace transform.

Inverse Laplace transform

Theorem

- Let f be a continuous function on $[0, \infty)$ that has a Laplace transform, $\mathscr{L}\{f\} = F$
- Then the *inverse Laplace transform* also exists, i.e.,

$$\mathscr{L}^{-1}\{F\} = f \quad \text{for all } t \in [0, \infty)$$

The inverse Laplace transform is linear:

1. $\mathscr{L}^{-1}\{F + G\} = \mathscr{L}^{-1}\{F\} + \mathscr{L}^{-1}\{G\} = f + g$
2. $\mathscr{L}^{-1}\{\beta F\} = \beta \mathscr{L}^{-1}\{F\} = \beta\, f$ where β is a constant

*A formula for the inverse exists but it requires advanced complex analysis.

Discontinuous functions

1. If f is discontinuous at $t = 0$, the inverse returns the limit from above, i.e.,

$$\mathscr{L}^{-1}\{F\}\Big|_{t=0} = \lim_{t\to 0^+} f(t)$$

2. If f is discontinuous at $\tau > 0$, the inverse returns the mid-point

$$\mathscr{L}^{-1}\{F\}\Big|_{t=\tau} = \frac{1}{2}\left(\lim_{t\to \tau^-} f(t) + \lim_{t\to \tau^+} f(t)\right)$$

MAST20030 Differential Equations Formulae Sheet

1) Laplace Transforms

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$ (Definition of Transform)
2.	1	$\frac{1}{s}$
3.	t^n	$\frac{n!}{s^{n+1}}$
4.	e^{at}	$\frac{1}{s-a}$
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

Exercise 3.7: Find the following Laplace transforms

(i) $2\mathrm{e}^{-3t}$ (ii) $\sin(2t) + \cos(2t)$

By linearity,

$$\mathscr{L}\{2\mathrm{e}^{-3t}\} = 2\,\mathscr{L}\{\mathrm{e}^{-3t}\}$$

$$= \frac{2}{s+3} \quad \text{from LT sheet}$$

for $\mathrm{Re}(s) > -3$

Again, using linearity

$$\mathscr{L}\{\sin(2t) + \cos(2t)\} = \mathscr{L}\{\sin(2t)\} + \mathscr{L}\{\cos(2t)\}$$

$$= \frac{2}{s^2+4} + \frac{s}{s^2+4} = \frac{s+2}{s^2+4}$$

for $\mathrm{Re}(s) > 0$

Exercise 3.8: Find the following inverse Laplace transforms

(i) $\dfrac{2}{s^4}$ (ii) $\dfrac{2}{s-1}+\dfrac{3s}{s^2+9}$

$$\mathscr{L}^{-1}\left\{\frac{2}{s^4}\right\}=\frac{1}{3}\mathscr{L}^{-1}\left\{\frac{3!}{s^4}\right\} \quad \text{by linearity}$$

$$=\frac{t^3}{3} \quad \text{from LT sheet}$$

$$\mathscr{L}^{-1}\left\{\frac{2}{s-1}+\frac{3s}{s^2+9}\right\}=2\,\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\}+3\,\mathscr{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\}$$

$$=2\,\mathrm{e}^t+3\,\cos(3t)$$

Partial fractions; **Exercise 3.9**: Find the inverse Laplace transform of

$$F(s) = \frac{2}{(s-1)(s-2)}$$

Seek A and B such that

$$F(s) = \frac{A}{s-1} + \frac{B}{s-2} = \frac{A(s-2)+B(s-1)}{(s-1)(s-2)}$$

$$\Rightarrow \quad 2 = A(s-2)+B(s-1) = (A+B)\,s - (2A+B)$$

$$\Rightarrow \quad A+B=0 \quad \text{and} \quad 2A+B=-2$$

$$\Rightarrow \quad A=-B \quad \Rightarrow \quad 2A-A=A=-2 \quad \Rightarrow \quad B=2$$

Thus

$$\mathscr{L}^{-1}\{F(s)\} = \mathscr{L}^{-1}\left\{\frac{-2}{s-1}+\frac{2}{s-2}\right\} = -2\,\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} + 2\,\mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\}$$